

OXIEDO: The Architecture That Unlocks Locked Data

The most valuable data in the world cannot be trained on, because no model can account for what it did with it. OR-

383 Controlled Experiments Review the Natural PS 2026 Workshop it learns. Submitted to AAAI 2027

| One Decision in 1986. Three Problems Everyone Funds Separately.

A Single Global Error Signal Makes Fault Localization Mathematically Impossible — Every Downstream Failure Follows From It

NOTHING CAN BE LOCATED

One error signal spreads across every parameter at once.

No unit can assess its own state.

The loss curve collapses and nothing says which node died.

NOTHING IS RETAINED

New tasks overwrite prior representations indiscriminately.

ResNet-18 keeps 47.3% after a sequential task shift.

Every regime change costs what the model knew.

NOTHING IS REPAIRED

Repair requires knowing what is broken.

Corrupted data is learned silently and confidently.

Rollback or full retrain are the only recourse.

One cause: the chain rule multiplies every unit's contribution through every other unit's. Unchanged since 1986.

| Kill a Converged Layer. One Architecture Comes Back.

Bounding Each Node's Local Gradient Chain Produces Attribution as a Physical Byproduct — and Attribution Makes Repair Possible

STANDARD BACKPROPAGATION

One global error signal, entangled across all parameters.

- No per-node visibility.
- Silent collapse on node death.
- No diagnosis, no targeted repair.

Feature layer zeroed at epoch 100:

10.0% — chance, every seed, zero variance, permanent.

ORMAS THREE-SIGNAL TRAINING

Each node reads through a 4,416-parameter bottleneck — too small to memorise noise.

- Scores its own health every step.
- Fires at 2.5σ below its own history.
- Diagnoses 7 modes, repairs mid-run.

Same attack: 80.3% after 85 targeted corrections (+70.3pp).

Local stability characterization from Input-to-State Stability (Sontag, 1989) — not a global convergence proof, not yet peer-reviewed.

| The Industry Is Moving From Inference to Training on Private Data

Three Independent Pressures Converge on the Same Missing Layer — None of Them Existed Two Years Ago

REGULATORS SHIFTED TO ARCHITECTURE

The FDA rejects medical models for opacity **regardless of accuracy**.

The EU AI Act attaches liability to systems with no audit trail.

The requirement is no longer performance. It is provenance.

RENTED INFERENCE STOPPED SCALING

Per-call cost compounds linearly with usage.

At institutional volume the only exit is your own infrastructure.

Which means training on your own data — the thing nobody can do safely.

MODELS DEGRADE INVISIBLY

Every regime change, patient cohort and fraud pattern erodes what the model knew.

No detection exists at the training layer.

It surfaces as a miscalculation, not an alert.

The regime is shifting from inference on public data to training on proprietary institutional data.

| Five Sectors. One Question. Counted Up, Not Down.

Every Sector Holding Data Has This Problem — These Five Are Where the Missing Account Is Already Costing Somebody Money in Figures That Can Be Checked

THE QUALIFYING BUYER

Three properties at once, or not a customer:

1. Holds data that legally cannot leave its infrastructure.
2. Has an active initiative to train on it.
3. Answers to somebody who will ask what the model learned.

The constraint makes the count small and the contracts large.

WHERE THE BILL ALREADY ARRIVES

AI Training — \$16K–\$24K per two-hour recovery from a run that degraded silently.

Medical AI — 12–36 months added to market entry when an audit asks what training changed.

Regulated Finance — a model that cannot be audited component by component does not clear validation. The programme is written off.

Data Obligation — \$100K–\$500K per deletion request, because retraining from scratch is the only compliant remedy.

Defense & Safety-Critical — “it modified itself and we cannot say how” ends the deployment.

| One Product. Three Things Arrive With the Model.

Training Is Unchanged — Data, Objective and Resulting Model Are Identical. What Changes Is the Account That Exists Alongside It

01 — THE MODEL

Deploys exactly as it would have

Same architecture, same inference cost, no change to the serving path.

Nothing is traded away for the accountability. That is the whole point.

02 — THE NAMED PARTS

What it learned, per source

Each source's contribution held as a discrete addressable object, bound to the record of the data that justified it.

Removable on instruction, without retraining.

03 — THE RECORD

Every change training made

Which component, at which step, under which diagnosis, by how much, inside what declared limit.

Signed, and diffable against the last approved version.

Seven features of one architecture, not seven products. Nothing is priced per feature. [anonymous.4open.science/r/ormas-EB73](https://www.reddit.com/r/openai/comments/10ormas/anonymous_4open_science_rormas_eb73/) — all 383 runs reproduce with one command.

I Registered a Company to Try to Disprove My Own Architecture

Seven Escalating Tests Over Twelve Months, Each Designed to Fail in Public — All Seven Held

RUNGS 1–4

1. **Origination** — demand from nothing? Cold start, no brand, no ad budget. **Customers arrived, 100% LLM-referred.**
2. **Unit economics** — **\$0.0043** per 12-asset suite vs \$50–\$150 human.
3. **Causal isolation** — was it the AI or the market? System stripped, all else identical. **–91% output.**
4. **Collapse morphology** — **–100% new customers**, all channels at once.

RUNGS 5–7

5. **Reversibility** — lesion or permanent? Re-injected, nothing else changed. **Immediate recovery.**
 6. **Recovery magnitude** — **+1,300%**, overshooting **3.3×** on accumulated memory alone.
 7. **Transaction ceiling** — can it close something expensive? **\$4,386** order at **\$0.00** CAC, no human involved.
- The scale stops there because I stopped it** — no jurisdiction assigns liability to an autonomous agent.

What this is evidence for, precisely: the orchestration layer that ran the company — prior work, now closed. It is not evidence for ORMAS, which carries its own separate experimental record. Total revenue across twelve months was \$6,691.68 on 79+ orders; the study was capped at research grade deliberately.

| Publish the Mechanism. License the Accountability.

Defensibility Never Rested on Secrecy — It Rests on Calibration, Regulatory Acceptance, and a Record a Reimplementation Does Not Come With

PUBLISHED — FREE

The architecture and every experiment behind it release in full on publication.

Free for research, teaching and evaluation, permanently.

A result nobody can check is worth nothing to a regulator.

Written into the licence, not promised in a blog post.

LICENSED — REVENUE

Production deployment, on the customer's own infrastructure.

Scope agreed per institution and written into the contract.

One integration, one contract, nothing priced per feature.

Zero data leaves the perimeter, so margin is software-grade.

WHAT IS ACTUALLY BOUGHT

Not the architecture — that is free from release day.

A calibrated bound, and a counterparty who will stand behind it in a filing.

Calibration is accumulated runs, adversarial testing and failure history.

A reimplementation arrives with none of it.

| The Regulator Creates the Demand. A Person Has to Close It.

Compliance Deadlines Generate Inbound Pull — the Remaining Gap Is Someone Who Can Sit in the Procurement Meeting

WHY THEY COME TO US

The buyer already has the problem written down by someone else.

- Regulators reject models on architecture, not accuracy — the requirement is documented before we arrive.
- The audit trail is produced during training, not reconstructed after.
- Nothing leaves the client's infrastructure — removing the objection that kills every SaaS alternative.

THE ONE GAP

The system replaced every operational role in a company, with causal data on how much of it was the system.

It cannot sit in a compliance review and be a person the buyer holds accountable.

Regulated institutions buy from someone who has survived their procurement process.

That is the co-founder. Not an engineer, not a generalist.

Everyone Is Competing One Layer Above the Constraint

Orchestration Is Not the Blocker — the Blocker Is That You Cannot Safely Train on the Data

	Per-node attribution	Stability analysis	Autonomous repair	Training-time audit trail	Runs on their metal
Agent frameworks	×	×	×	×	×
Enterprise data platforms	×	×	×	~	~
Frontier model vendors	×	×	×	×	×
Data labelling vendors	×	×	×	×	×
Academic research	~	~	~	×	—
This	✓	✓	✓	✓	✓

The moat is not secrecy — the work is published. It is that reaching this required breaking a live company on its own data first, then control theory, then two years of experiments. That order does not compress.

The assets worth owning are the deployment record, the accumulated institutional memory, and a model nobody else has started.

| Everything Technical Is Built. One Thing Is Not.

Three Years Solo Produced the Architecture, the System and the Causal Proof — the Gap Is a Person, Not a Capability

FOUNDER

Eighteen. No university, no lab, no advisor, no external funding.

A self-correcting training architecture with a formal stability result, derived at 17.

An agent operating system that ran a live company for twelve months.

A UK entity registered from abroad at 17, unassisted, against an AML barrier.

Programme committee reviewer, Trustworthy AI for Good workshop at **NeurIPS 2026**.

Every experiment ran on one GPU, bought at 15 with money from the first thing he sold.

CO-FOUNDER SOUGHT

One specific profile:

Has sold software inside a regulated data environment — a fund, a hospital, an insurer.

Understands model risk well enough to argue with a compliance officer and win.

Not an engineer. Not a generalist. *Equal equity, four-year vest.*

| From a Training Layer to the Substrate Regulated AI Runs On

Every Phase Removes One Dependency — Ending With a Stack That Rents Intelligence From No One

PHASE 1 — NOW

Finish the research and carry per-node health monitoring onto attention heads.

Architecture-agnostic by design, so this is a compute problem, not a research one.

First design partners.

PHASE 2 — 12-24 MONTHS

First licensed deployments inside institutions that cannot move their data.

Each accumulates institutional memory that cannot be exported.

Switching cost compounds with tenure.

PHASE 3 — YEARS 3-5

A model pre-trained on the self-correcting architecture from scratch.

The external dependency disappears; cost collapses to hardware.

A position no competitor reaches by adding a feature.

The network learns to stop needing correction. That is the thesis, and it is measured.

Correction frequency decays $4.2 \rightarrow 0.05$ per epoch — the system repairs itself until it no longer needs to.

| \$1.5M Pre-Seed. Sized to the Next Result, Not the Whole Plan.

Eighteen Months, Six Milestones With Acceptance Criteria — and a Compute Line Costed Against the Question Actually Open, Not the One After It

USE OF FUNDS

Team — co-founder and a research engineer
\$650K

Compute — 32–64 accelerators, a 3–4 week campaign, run twice
\$250K

Dataset partnership — one custodian, non-exclusive
\$150K

Structure and filings — Delaware, EU representation, priority filings
\$100K

Operating — 18 months of everything else
\$200K

Buffer — unallocated, on purpose
\$150K

WHAT IT BUYS, IN ORDER

1. **A second and third person.** Key-person risk is total today; this is the first thing the money fixes.

2. **One result that is not CIFAR.** Bounded attribution demonstrated on a transformer backbone, on the same adversarial suite the current architecture was hardened against.

3. **One custodian saying yes.** No clinical, financial or defence data has ever touched this system. Compute does not fix that; a partnership does.

The frontier-scale campaign is a seed-round problem. It is not on this page, and it is not being asked for here.